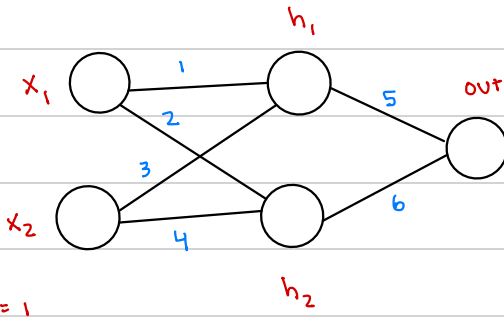


Forward propagation / Forward pass

Neural network:



Activation function:

Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$x_1 = 1$$

$$x_2 = 0.5$$

Edge weights:

1. 0.1

2. 0.3

3. 0.2

4. 0.4

5. 0.5

6. 0.6

Forward to:

$$h_1: 1 \cdot 0.1 + 0.5 \cdot 0.2 = 0.2$$

$$h_2: 1 \cdot 0.3 + 0.5 \cdot 0.4 = 0.5$$

$$\text{out}: 0.550 \cdot 0.5 + 0.622 \cdot 0.6 = 0.648$$

Out of:

$$h_1: \sigma(0.2) = 0.550$$

$$h_2: \sigma(0.5) = 0.622$$

$$\text{out}: \sigma(0.648) = \boxed{0.657}$$

○ → out = +1 if out ≥ 0
-1 otherwise